CSP and Specht Modules

Directed Reading Project Reference List and 6-Week Study Plans

Includes the original fuller plan and a revised 3-5 hours/week version

Project framing

This project investigates the cyclic sieving phenomenon (CSP), a connection between cyclic group actions, q-analogues, and combinatorics. In its simplest form, CSP asserts that evaluating a q-polynomial at roots of unity counts fixed points under a cyclic symmetry. The project studies rotating lattice paths, promotion on standard Young tableaux, descent statistics, and the representation-theoretic interpretation through Specht modules for the symmetric group.

Guiding question: Why do root-of-unity evaluations of natural q-analogues know fixed-point counts for cyclic actions?

Reference list

The first four items are the core references. The remaining items are useful background or backup sources depending on student preparation.

- **Reiner, Stanton, White, "The Cyclic Sieving Phenomenon", JCTA, 2004.** Original CSP paper. Use for the definition and foundational examples: q-binomial coefficients, q-hook formulas, Schur functions, and root-of-unity evaluations.
- **Bruce Sagan, "The Cyclic Sieving Phenomenon: A Survey", 2010.** Readable overview. Good for examples and for seeing common proof patterns behind CSP.
- **Brendon Rhoades, "Cyclic Sieving, Promotion, and Representation Theory", JCTA, 2010.** Central paper for promotion on rectangular standard Young tableaux and the representation-theoretic explanation.
- **Bruce Sagan, The Symmetric Group: Representations, Combinatorial Algorithms, and Symmetric Functions.** Best single book for Young tableaux, Specht modules, symmetric group representations, and the combinatorial representation theory needed here.

Additional background references

- **William Fulton, Young Tableaux.** Concise source for Young diagrams, tableaux, Schur functions, and the representation theory of S_n and GL_n.
- **Richard Stanley, Enumerative Combinatorics, Volume 2.** Useful for symmetric functions, hook-length formulas, principal specializations, P-partitions, and q-hook-length formulas.
- **Gordon James, The Representation Theory of the Symmetric Groups.** More algebraic backup for tabloids, polytabloids, Specht modules, and standard bases.
- **SageMath combinatorics documentation on cyclic sieving.** Useful for computational experiments with q-binomial coefficients and roots of unity.

Original 6-week study plan

> *This is the fuller version of the plan: ambitious, conceptually complete, and suitable for students who can spend more time outside meetings.*

Week 1 - What is cyclic sieving?

Goal: Understand the definition of CSP and see why it is surprising.

Reading: Read the introduction and first examples from Reiner-Stanton-White, and the definition/examples from Sagan survey.

Activity/exercise: Compute fixed points of k-subsets of [n] under cyclic rotation. For n=4 and k=2, compare the fixed-point counts with evaluations of the Gaussian binomial coefficient [4 choose 2]_q at 1, i, -1, and -i.

Deliverable/outcome: Each student can explain one CSP example in the form: fixed points under symmetry = root-of-unity evaluation of a q-analogue.

Week 2 - Young diagrams, tableaux, descents, and major index

Goal: Learn the tableau combinatorics needed for the representation-theoretic side.

Reading: Read Fulton or Sagan on Young diagrams/tableaux, then read a short section on descents and major index.

Activity/exercise: Work out all standard Young tableaux of shapes (2,1), (2,2), and (3,1). Compute the descent set Des(T), the major index maj(T), and the polynomial sum_T q^maj(T).

Deliverable/outcome: A one-page table listing the tableaux, descent sets, and major indices for the small shapes.

Week 3 - Specht modules and the symmetric group

Goal: Understand how tableaux become representations of S_n.

Reading: Read Sagan on tabloids, polytabloids, permutation modules, and Specht modules. Use Fulton or James only as backup.

Activity/exercise: Construct S^(2,1) explicitly for S_3. Compare the irreducibles indexed by (3), (2,1), and (1,1,1), whose dimensions are 1, 2, and 1.

Deliverable/outcome: Students can explain why dim S^lambda = number of standard Young tableaux of shape lambda.

Week 4 - Promotion, evacuation, and cyclic actions on tableaux

Goal: Introduce the cyclic action that produces CSP on tableaux.

Reading: Read the promotion sections in Sagan survey and the introduction of Rhoades.

Activity/exercise: Define Schutzenberger promotion: remove 1, slide by jeu de taquin, decrement labels, and place n in the final empty box. Compute promotion orbits for small rectangular shapes, especially (2,2).

Deliverable/outcome: A small worked example verifying the promotion CSP for a rectangular tableau shape by hand.

Week 5 - Characters, fake degrees, and why representation theory explains CSP

Goal: Explain the mechanism behind CSP: root-of-unity evaluations behave like character evaluations.

Reading: Read the representation-theoretic discussion in Rhoades and Sagan survey. Review character/traces from Sagan if needed.

Activity/exercise: Discuss the chain: major-index polynomial -> graded multiplicity/fake degree -> character value -> trace -> fixed-point count. Use the principle that if an operator permutes a basis, its trace equals the number of basis elements fixed by the operator.

Deliverable/outcome: Students prepare a diagram connecting SYT(lambda), S^lambda, f^lambda(q), root-of-unity evaluations, and promotion fixed points.

Week 6 - Final synthesis and presentations

Goal: Put the whole CSP/tableaux/Specht module story together.

Reading: Review the relevant parts of the first five weeks and revisit one example in detail.

Activity/exercise: Students give short presentations or write expository notes. Possible topics: rotating lattice paths, q-hook formula, Specht modules, promotion, major index, fake degrees, or why CSP is not a coincidence.

Deliverable/outcome: A 10-15 minute presentation or short note answering: How do Specht modules help explain CSP for standard Young tableaux?

Conceptual spine: q-binomial coefficients -> major-index polynomial f^lambda(q) -> Specht module S^lambda -> promotion CSP.

Revised 6-week plan with 3-5 hours per week

> *This version is designed for a realistic DRP workload: about 1 hour for the weekly meeting plus 2-4 hours of preparation, exercises, and reflection.*

Week 1 - CSP from rotating objects

Workload: 3-4 hours total: 1.5-2 hours reading, 1 hour meeting, 0.5-1 hour exercise.

Goal: State the CSP definition and understand one complete example.

Reading: Sagan survey Sections 1-2, or equivalent first examples. Skim the introduction of Reiner-Stanton-White.

Activity/exercise: Verify CSP for 2-subsets of [4] under rotation using [4 choose 2]_q. If time remains, repeat for binary words with two 1s.

Deliverable/outcome: Student can state the definition and explain one example without relying on the notes.

Week 2 - Tableaux, descents, and major index

Workload: 3-5 hours total: 1.5-2.5 hours reading, 1 hour meeting, 0.5-1.5 hours exercises.

Goal: Understand how a q-polynomial can be built from a statistic on tableaux.

Reading: Short section from Sagan or Fulton on Young diagrams and standard Young tableaux. Short notes/section on descent set and major index.

Activity/exercise: List all SYT of shapes (2,1) and (2,2). Compute Des(T), maj(T), and sum_T q^maj(T).

Deliverable/outcome: Student can explain descent set and major index in examples.

Week 3 - Hook formulas and q-hook formulas

Workload: 3-5 hours total: 1.5-2 hours reading, 1 hour meeting, 0.5-2 hours computations.

Goal: See f^lambda(q) as a q-analogue of the dimension/count f^lambda.

Reading: Read about hook lengths and the hook-length formula. Skim the q-hook/major-index formula without requiring a full proof.

Activity/exercise: Compute hook lengths and f^lambda for (3,1), (2,2), and (2,1,1). Compare the q=1 value of the q-hook expression with the ordinary hook formula.

Deliverable/outcome: Student understands the relationship between ordinary counting and q-counting for tableaux.

Week 4 - Specht modules without too much algebra

Workload: 4-5 hours total: 2-2.5 hours reading, 1 hour meeting, 1-1.5 hours example writing.

Goal: Understand the slogan: tableaux index bases for irreducible S_n-representations.

Reading: Read Sagan on tabloids, polytabloids, and Specht modules, focusing on examples rather than all proofs.

Activity/exercise: Construct the representations indexed by (3), (2,1), and (1,1,1) for S_3. Explain why dim S^lambda = number of SYT(lambda).

Deliverable/outcome: Student can describe how Specht modules connect tableaux to representation theory.

Week 5 - Promotion and tableau CSP

Workload: 3-5 hours total: 1.5-2 hours reading, 1 hour meeting, 0.5-2 hours computation.

Goal: Explain the rectangular-tableau CSP example concretely.

Reading: Read the promotion/CSP section of Sagan survey and the introduction of Rhoades. Do not attempt the full proof unless the group is advanced.

Activity/exercise: Compute promotion orbits for SYT of shape (2,2). Evaluate the major-index polynomial at the relevant roots of unity and compare with fixed tableaux.

Deliverable/outcome: Student can connect promotion, fixed points, and root-of-unity evaluations in one example.

Week 6 - Why representation theory explains CSP

Workload: 3-4 hours total: 1 hour review, 1 hour meeting, 1-2 hours final preparation.

Goal: Connect promotion, fixed points, q-hook/major index, and Specht modules in one coherent story.

Reading: Light reading on characters, traces, and fake degrees from Sagan survey or Rhoades. Focus on the trace = fixed points principle.

Activity/exercise: Prepare a 5-8 minute mini-presentation or 1-2 page note: "How do Specht modules help explain CSP?"

Deliverable/outcome: Student can give the final conceptual map of the project.

Recommended weekly rhythm

Time block	Task
Before meeting: 1.5-2.5 hours	Read assigned pages and write down definitions or questions.
During meeting: 1 hour	Discuss the main example and work through one computation together.
After meeting: 0.5-1.5 hours	Finish the short exercise and write a few sentences summarizing the week.

Minimal path versus stretch path

Week	3-hour version	5-hour version
1	One q-binomial CSP example.	Compare k-subsets with lattice paths or binary words.
2	Descents and major index for (2,1) and (2,2).	Add (3,1) and compare with a q-hook formula.
3	Ordinary hook-length formula.	Verify the q-hook formula in small examples.
4	Specht modules for S_3 only.	Discuss S_4 examples or a small character table.
5	Promotion on (2,2).	Promotion on (3,2), possibly with Sage/computation.
6	Final conceptual map.	Connect to fake degrees or graded representations.

Final deliverable options

- A 5-8 minute presentation explaining one CSP example from start to finish.
- A 1-2 page expository note on how major index and Specht modules enter the CSP story.
- A computational notebook checking CSP for small q-binomial or tableau examples.
- A final concept map connecting cyclic action, fixed points, q-polynomial, root-of-unity evaluation, tableaux, major index, and Specht modules.

Suggested final synthesis statement

Tableaux encode irreducible representations of S_n through Specht modules. Statistics such as major index produce q-analogues of dimensions. Promotion gives a cyclic symmetry on tableaux. In favorable cases, evaluating the major-index/q-hook polynomial at roots of unity recovers fixed-point counts for promotion. This is the cyclic sieving phenomenon.